

AVCDL NHTSA Cybersecurity Best Practices Fulfillment

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Abstract

This document describes how **AVCDL** phase requirement products fulfill the **NHTSA Cybersecurity Best Practices for Safety of Modern Vehicles** recommendations.

Motivation

This document is motivated by the need to justify the sufficiency of the **AVCDL** for compliance with the **NHTSA Cybersecurity Best Practices for Safety of Modern Vehicles**.

Audience

The audience for this document is the certifying organization.

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Note Regarding NHTSA Guidance

The **NHTSA Cybersecurity Best Practices for Safety of Modern Vehicles** is a voluntary guidance document. It is targeted at both individual organizations and the automotive industry as a whole. As the document states:

NHTSA has made vehicle cybersecurity an organizational priority, and it is important for automotive industry suppliers and manufacturers to do so as well.

and

NHTSA believes the voluntary best practices described in this document provide a solid foundation for developing a risk-based approach to cybersecurity challenges, and describes important processes that can be maintained, refreshed and updated effectively over time to serve the needs of the automotive industry.

The document does not restrict its recommendations to product cybersecurity. Because of this it must be considered by a host of entities within and without an organization, as well as across the supply chain.

This document is intended to cover cybersecurity issues for all motor vehicles and motor vehicle equipment (including software) and is therefore applicable to all individuals and organizations designing and manufacturing vehicle electronic systems and software. These entities include, but are not limited to, small- and large-volume motor vehicle and motor vehicle equipment designers, suppliers, manufacturers, modifiers, and alterers.

Additionally, given the stated focus on safety, many aspects of product cybersecurity engineering are excluded in preferences to a safety risk centric set of recommendations.

Prioritizing vehicle cybersecurity also means establishing internal processes and strategies to ensure systems will be safe under expected real-world conditions, including in the presence of potential vehicle cybersecurity threats.

Application of Information

Within the context of the **NHTSA Cybersecurity Best Practices for Safety of Modern Vehicles**, as presented in this document, the **AVCDL** will be shown where it either directly implements or supports the various practices identified by **NHTSA**. It is expected that any organization seeking to apply the guidance within the **NHTSA Cybersecurity Best Practices for Safety of Modern Vehicles** document will have a compliance group leading the interaction with **NHTSA** and coordinating with the product cybersecurity group regarding those aspects requiring that group's support.

Furthermore, given the scope of the full lifecycle supported by the **AVCDL**, it is expected that any implementer of the **AVCDL** or authority external to the organization fully review the material in context. This is because there is an explicit traceability established by the **AVCDL** which removes the necessity to reiterate every precursor activity. This traceability is shown in the **AVCDL Traceability** elaboration document.

It is assumed that since the specific practices identified by **NHTSA** which an organization undertakes will be governed to a large extent by their position within the supply chain, and that the application of the **AVCDL** may be in support of the overall activities of the manufacturer and/or any organization within the supply chain providing components for the vehicle under consideration.

Additionally, it is presumed that during organizational review activities toward **NHTSA** approval that individuals competent to explain and defend the processes expressed in and products generated by the application of the **AVCDL** will be available to that authority's examiners.

In addition to the **AVCDL** primary and secondary documents, the **AVCDL** elaboration documents provide contextual information which may be helpful to the approval authority.

Construction of Argument

As noted previously, this document uses **NHTSA Cybersecurity Best Practices for Safety of Modern Vehicles** (Updated 2022) as the basis for determining the **AVCDL** material relevant to each practice enumerated within that document.

Also utilized are the **NHTSA cybersecurity best practices** workbook spreadsheets. The purpose of these is purely reference (primarily for the creation of other materials).

The **AVCDL NHTSA Cybersecurity Best Practices Fulfillment** document is the focus for guiding any review, not the **NHTSA cybersecurity best practices** workbook spreadsheets.

Note: The **NHTSA cybersecurity best practices** workbook spreadsheets are only a tool used to enable the construction of the argument presented within this document. They are referenced only for the purpose of explanation and are not expected to be necessary to understanding how the **AVCDL** fulfills various **NHTSA Cybersecurity Best Practices for Safety of Modern Vehicles** identified practices.

NHTSA Cybersecurity Best Practices Overview

Note: This material is extracted from the **NHTSA Cybersecurity Best Practices for Safety of Modern Vehicles** document. It is included here for reference only.

NHTSA Cybersecurity Best Practices for Safety of Modern Vehicles presents recommendations regarding cybersecurity practices applicable, within the context of safety, to modern road vehicles.

That document contains the following sections:

- [General Cybersecurity Best Practices](#) (4)
- [Education](#) (5)
- [Aftermarket / User-Owned Devices](#) (6)
- [Serviceability](#) (7)
- [Technical Vehicle Cybersecurity Best Practices](#) (8)

Note: Out-of-scope activities are addressed by the noted non-cybersecurity group.

4 General Cybersecurity Best Practices

4.0 General Cybersecurity Best Practices

[G.1] The automotive industry should follow the National Institute of Standards and Technology's (NIST's) documented Cybersecurity Framework, which is structured around the five principal functions, "Identify, Protect, Detect, Respond, and Recover," to build a comprehensive and systematic approach to developing layered cybersecurity protections for vehicles.

Note: This recommendation is addressed by the IT and OT groups.

References

Framework for Improving Critical Infrastructure Cybersecurity

<https://doi.org/10.6028/NIST.CSWP.04162018>

Discussion

The **AVCDL** does not address this recommendation as it does not fall within the context of product cybersecurity. As seen from the title of the reference document, this recommendation is framed within the context of infrastructure.

4.1 Leadership Priority on Product Cybersecurity

[G.2] Companies developing or integrating vehicle electronic systems or software should prioritize vehicle cybersecurity and demonstrate executive management commitment and accountability by:

Note: This recommendation is addressed in the organizational-level documentation.

References

ISO/SAE 21434 [WP-05-01][RQ-05-01] cybersecurity policy

<https://www.iso.org/standard/70918.html>

[a] Allocating dedicated resources within the organization focused on researching, investigating, implementing, testing, and validating product cybersecurity measures and vulnerabilities;

Discussion

The **AVCDL** does not address this recommendation as the establishment of cybersecurity policy is done at the organizational-level.

[b] Facilitating seamless and direct communication channels through organizational ranks related to product cybersecurity matters;

Note: This recommendation is addressed in the organizational-level documentation.

Discussion

Although the **AVCDL** does not directly address this work product, as it is an organization-level responsibility, there is useful discussion in the **AVCDL** primary document. **Communication channels** are addressed in the **AVCDL** primary document section 3.3 – **Inter-group Communication**. **Information sharing** is addressed in the **AVCDL** primary document section 3.2 – **Information Sharing**.

[c] Enabling an independent voice for vehicle cybersecurity-related considerations within the vehicle safety design process.

Note: This recommendation is addressed in the organizational-level documentation.

Discussion

The **AVCDL** does not address this recommendation as the independence of product cybersecurity is established and ensured at the organizational-level.

4.2 Vehicle Development Process With Explicit Cybersecurity Considerations

4.2.1 Process

[G.3] The automotive industry should follow a robust product development process based on a systems-engineering approach with the goal of designing systems free of unreasonable safety risks, including those from potential cybersecurity threats and vulnerabilities.

Note: This recommendation is partially supported.

Discussion

Although the **AVCDL** itself forms the basis for a **cybersecurity development lifecycle**, it depends upon the existence of both foundational development lifecycles as described in **ISO/IEC 15288 (system development lifecycle)** and **ISO/IEC/IEEE 12207 (software development lifecycle)**. These being the responsibility of the systems engineering and development groups, respectively. Additionally, safety concerns are outside the scope of the product cybersecurity group.

4.2.2 Risk Assessment

[G.4] This process should include a cybersecurity risk assessment step that is appropriate and reflects mitigation of risk for the full lifecycle of the vehicle.

References

ISO/SAE 21434 [WP-09-02][RQ-09-03] item analysis

ISO/SAE 21434 [WP-09-02][RQ-09-04] risk treatment

Discussion

The **AVCDL** provides not only a specific set of processes for handling design-level analysis, but also a generalized approach for risk treatment (threat prioritization). Additionally, the **Understanding TARA in an AVCDL Context** elaboration document provides additional context as to the **AVCDL** threat analysis approach.

[Foundation-9] Threat Prioritization Plan (Foundation-9.1)

[Design-4] Threat Modeling Report (Design-4.1)

[Design-4] Ranked / Risked Threat Report (Design-4.2)

[Design-4] Threat Report (Design-4.3)

[G.5] Safety of vehicle occupants and other road users should be of primary consideration when assessing risks.

Note: This recommendation is partially supported.

Discussion

The **AVCDL** does not directly address the specific domain of risk considered as that is the responsibility of the organization-level risk management group. An extensive discussion of risk and risk domains is provided in the **AVCDL** primary document section 3.4.4 **Risk**.

[Design-4] Ranked / Risked Threat Report (Design-4.2)

4.2.3 Sensor Vulnerability Risks

[G.6] Manufacturers should consider the risks associated with sensor vulnerabilities and potential sensor signal manipulation efforts such as GPS spoofing, road sign modification, Lidar/Radar jamming and spoofing, camera blinding, and excitation of machine learning false positives.

Discussion

The **AVCDL** does not consider individual compromise vectors, but rather failure to ensure identified cybersecurity properties. Additional information regarding cybersecurity properties can be found in the **Security Requirements Taxonomy** secondary document. All requirement and design phase activities treat all elements within the system uniformly.

[Requirements-1] Element-level Security Goals (Requirements-1.1)

[Requirements-1] Element-level Security Requirements (Requirements-1.2)

[Design-1] Design Showing Security Considerations (Design-1.1)

[Design-2] Security Design Review (Design-2.1)

[Design-4] Attack Surface Analysis Report (Design-3.1)

[Design-4] Threat Modeling Report (Design-4.1)

[Design-4] Ranked / Risked Threat Report (Design-4.2)

[Design-4] Threat Report (Design-4.3)

4.2.4 Removal or Mitigation of Safety-Critical Risks

[G.7] Any unreasonable risk to safety-critical systems should be removed or mitigated to acceptable levels through design, and any functionality that presents an unavoidable and unnecessary risk should be eliminated where possible.

Discussion

The **AVCDL** presumes that identified issues prioritized as requiring mitigation are addressed at the proper phase within the lifecycle. Additional information regarding process feedback can be found in the **Understanding Cybersecurity Risk Freshness in an AVCDL Context** elaboration document.

[Foundation-9] Threat Prioritization Plan (Foundation-9.1)

[Design-1] Design Showing Security Considerations (Design-1.1)

[Design-2] Security Design Review (Design-2.1)

[Design-4] Attack Surface Analysis Report (Design-3.1)

[Design-4] Threat Modeling Report (Design-4.1)

[Design-4] Ranked / Risked Threat Report (Design-4.2)

[Design-4] Threat Report (Design-4.3)

4.2.5 Protections

[G.8] For remaining functionality and underlying risks, layers of protection that are appropriate for the assessed risks should be designed and implemented.

Discussion

The **AVCDL** implementation phase activities are constructed to ensure that an appropriate level of cybersecurity engineering rigor is applied. Additional information regarding this methodology is covered in the **Cybersecurity Engineering Approach** secondary document.

[Implementation-1]	List of Tools and Components Used (Implementation-1.1)
[Implementation-2]	Build Process Documentation (Implementation-2.1)
[Implementation-3]	Secure Settings Document (Implementation-3.1)
[Implementation-4]	Component/Version – Product/Version Cross-reference Document (Implementation-4.1)
[Implementation-5]	Secure Development (Implementation-5.1)
[Implementation-6]	Currently Used Deprecated Functions Document (Implementation-6.1)
[Implementation-7]	Static Analysis Report (Implementation-7.1)
[Implementation-8]	Dynamic Analysis Report (Implementation-8.1)
[Implementation-9]	Secure Code Review Summary (Implementation-9.1)
[Implementation-10]	Fuzz Testing Report (Implementation-10.1)

[G.9] Clear cybersecurity expectations should be specified and communicated to the suppliers that support the intended protections.

References

ISO/SAE 21434 clause 7 Distributed Cybersecurity Activities

Discussion

The **AVCDL** provides a set of materials and processes for addressing the supply chain. Additional information is provided in the **Understanding Service Level Agreements in an AVCDL Context**, **Understanding Supplier Cybersecurity Process Mapping**, and **Understanding Supply Chain Interaction in an AVCDL Context** elaboration documents.

[Supplier-1]	AVCMDS (Supplier-1.1)
[Supplier-2]	Supplier Self-reported Maturity (Supplier-2.1)
[Supplier-3]	Cybersecurity Interface Agreement (Supplier-3.1)

4.2.6 Inventory and Management of Hardware and Software Assets on Vehicles

[G.10] Suppliers and vehicle manufacturers should maintain a database of their operational hardware and software components used in each automotive ECU, each assembled vehicle, and a history log of version updates applied over the vehicle's lifetime.

Note: This recommendation is partially supported.

References

Multistakeholder Process on Promoting Software Component Transparency, 83 Fed. Reg. 110 (June 4, 2018)

https://www.ntia.gov/sites/default/files/publications/fr-notice-07192018-meeting-software-component-transparency_0.pdf

Discussion

The **AVCDL** provides support and guidance to the overall software asset management process undertaken by the development group. Additional material can be found in the **Software Bill of Materials Lifecycle** and **Manifest Generation** elaboration documents.

[Foundation-3]	List of Approved Tools and Components (Foundation-3.1)
[Implementation-1]	List of Tools and Components Used (Implementation-1.1)
[Release -2]	Archive Manifest (Release -2.1)
[Operation-4]	Software Deployment Report (Operation-4.1)
[Decommissioning-1]	Decommissioning Report (Decommissioning-1.1)

[G.11] Manufacturers should track sufficient details related to software components, such that when a newly identified vulnerability is identified related to an open source or off-the-shelf software, manufacturers can quickly identify what ECUs and specific vehicles would be affected by it.

Note: This recommendation is partially supported.

Discussion

The **AVCDL** provides support and guidance to the overall software asset management process undertaken by the development group. Additional material can be found in the **Software Bill of Materials Lifecycle**, **Manifest Generation**, and **Understanding Open Source in an AVCDL Context** elaboration documents.

[Foundation-3]	List of Approved Tools and Components (Foundation-3.1)
[Foundation-5]	Code Protection Plan (Foundation-5.1)
[Foundation-7]	Incident Response Plan (Foundation-7.2)
[Implementation-1]	List of Tools and Components Used (Implementation-1.1)
[Release -2]	Archive Manifest (Release -2.1)
[Operation-1]	Cybersecurity Incident Report (Operation-1.1)
[Operation-2]	Cybersecurity Incident Report (Operation-1.1)
[Operation-3]	Cybersecurity Incident Report (Operation-1.1)
[Operation-4]	Software Deployment Report (Operation-4.1)
[Decommissioning-1]	Decommissioning Report (Decommissioning-1.1)

4.2.7 Cybersecurity Testing and Vulnerability Identification

[G.12] Manufacturers should evaluate all commercial off-the-shelf and open-source software components used in vehicle ECUs against known vulnerabilities.

References

MITRE Common Vulnerabilities and Exposures (CVE)

<https://cve.mitre.org/>

NIST's National Vulnerability Database

<https://nvd.nist.gov/>

ISO/SAE 21434 [WP-06-01] integration of off-the-self components discussion

ISO/SAE 21434 [WP-06-01][RQ-06-21] off-the-shelf component use

Discussion

The **AVCDL** provides mechanism to ensure the integrity of open source and third-party software ingested into the product. Additional material can be found in the **Understanding Open Source in an AVCDL Context** elaboration document.

[\[Foundation-3\]](#) [List of Approved Tools and Components \(Foundation-3.1\)](#)

[G.13] Manufacturers should also pursue product cybersecurity testing, including using penetration tests, as part of the development process.

References

ISO/SAE 21434 [WP-10-07][RC-10-12] verification of weakness minimization

Discussion

The **AVCDL** specifies a series of activities within both the implementation and verification phase to ensure code soundness. Additional material can be found in the **Understanding Verification and Validation in an AVCDL Context** elaboration document.

[\[Implementation-7\]](#) [Static Analysis Report \(Implementation-7.1\)](#)
[\[Implementation-8\]](#) [Dynamic Analysis Report \(Implementation-8.1\)](#)
[\[Implementation-9\]](#) [Secure Code Review Summary \(Implementation-9.1\)](#)
[\[Implementation-10\]](#) [Fuzz Testing Report \(Implementation-10.1\)](#)
[\[Verification-1\]](#) [Penetration Testing Report \(Verification-1.1\)](#)

[G.14] Test stages should employ qualified testers who have not been part of the development team, and who are highly incentivized to identify vulnerabilities.

Note: This recommendation is partially supported.

Discussion

The **AVCDL** only partially supports this requirement in that it is beyond the scope of product cybersecurity to ensure that resources are deployed such that they are not subject to the priorities of other groups. The **AVCDL** provides both a set of identified roles and responsibilities, and training appropriate for the activities to be undertaken within the scope of the **AVCDL**.

[Foundation-1] Training Catalog (Foundation-1.1)
[Foundation-1] System to Track Training Participation (Foundation-1.2)
[Foundation-2] Roles and Responsibilities Document (Foundation-2.1)

[G.15] A vulnerability analysis should be generated for each known vulnerability assessed or new vulnerability identified during cybersecurity testing. The disposition of the vulnerability and the rationale for the how the vulnerability is managed should also be documented.

References

ISO/SAE 21434 [WP-08-05] vulnerability analysis
ISO/SAE 21434 [WP-08-06][RQ-08-07] vulnerability management

Discussion

The **AVCDL** specifies use of a single process for prioritization of threat candidates. It also specifies the information to be contained within the various reports generated by the analysis activities.

[Foundation-9] Threat Prioritization Plan (Foundation-9.1)
[Implementation-7] Static Analysis Report (Implementation-7.1)
[Implementation-8] Dynamic Analysis Report (Implementation-8.1)
[Implementation-9] Secure Code Review Summary (Implementation-9.1)
[Implementation-10] Fuzz Testing Report (Implementation-10.1)
[Verification-1] Penetration Testing Report (Verification-1.1)

4.2.8 Monitoring, Containment, Remediation

[G.16] In addition to design protections, the automotive industry should establish rapid vehicle cybersecurity incident detection and remediation capabilities.

Note: This recommendation is partially supported.

Note: By definition recommendation directed toward the automotive industry are outside the scope of any individual organization.

References

ISO/SAE 21434 clause 13 Operations and Maintenance

Discussion

Incident detection is an ill-defined term in this context. The areas of IT and OT cybersecurity monitoring is outside the scope of the **AVCDL**. The area of cybersecurity monitoring within a vehicle is undefined territory. This falls outside the scope of the **AVCDL** as it is a part of the functional specification of the system which happens to provide cybersecurity-relevant value.

Incident response is an activity led by either the IT or OT group depending on the nature of the incident. As such, product cybersecurity provides support within this context. There are additional coordination elements should the supply chain be involved. The **AVCDL** addresses only those aspects directly related to product cybersecurity.

[Foundation-7]	Incident Response Plan (Foundation-7.2)
[Operation-1]	Cybersecurity Incident Report (Operation-1.1)
[Operation-2]	Cybersecurity Incident Report (Operation-1.1)
[Operation-3]	Cybersecurity Incident Report (Operation-1.1)

[G.17] Such capabilities should be able to mitigate safety risks to vehicle occupants and surrounding road users when a cyberattack is detected and transition the vehicle to a minimal risk condition, as appropriate for the identified risk.

Note: This recommendation is outside the scope of the **AVCDL**.

Discussion

Cybersecurity attack detection is discussed above as **incident detection** [G.16].

The activity of transitioning the vehicle to a minimal risk condition is outside the scope of the **AVCDL** as it is the purview of the safety group to establish the definitions of such, and the systems engineering group to design such an activity. These activities may be product cybersecurity adjacent. Where this is the case appropriate cybersecurity engineering support will be provided.

4.2.9 Data, Documentation, Information Sharing

[G.18] Manufacturers should collect information on potential attacks, and this information should be analyzed and shared with industry through the Auto-ISAC and other sharing mechanisms.

Note: This recommendation is partially supported.

References

ISO/SAE 21434 clause 8.3 Cybersecurity Monitoring

Discussion

Although the **AVCDL** has provisions for **cybersecurity monitoring**, participation in and information sharing with outside entities is an activity addressed at the organizational level. This is also true for the aspect of committing resources to perform analysis of potential attacks outside normal cybersecurity engineering activities.

[\[Foundation-7\]](#) [Cybersecurity Monitoring Plan \(Foundation-7.1\)](#)

[G.19] Manufacturers should fully document any actions, design choices, analyses, supporting evidence, and changes related to its management of vehicle cybersecurity.

Discussion

All **AVCDL** documentation both at the process and procedure level specifies that materials generated by cybersecurity activities be stored within the appropriate long term repository (QMS, RMS, or DMS). Additional information is provided in the **AVCDL** primary document section 8.6 **Availability of Products and Materials** and the **AVCDL Traceability** elaboration document.

[G.20] All related work products should be traceable within a robust document version control system.

Discussion

All **AVCDL** documentation both at the process and procedure level specifies that materials generated by cybersecurity activities be stored within the appropriate long term repository (QMS, RMS, or DMS). Additional information is provided in the **AVCDL** primary document section 8.6 **Availability of Products and Materials** and the **AVCDL Traceability** elaboration document.

Note: The capabilities, structure, and management of long-term repositories is not within the scope of the **AVCDL**.

4.2.10 Continuous Risk Monitoring and Assessment

[G.21] Companies should use a systematic and ongoing process to periodically reevaluate risks and make appropriate updates to processes and designs due to changes in the vehicle cybersecurity landscape, as appropriate.

References

ISO/SAE 21434 clause 8.3 Cybersecurity Monitoring

Discussion

The **AVCDL** addresses this recommendation in detail and scope well beyond the abovementioned reference. This information is provided in the **AVCDL** primary document sections 5 **Continuous Improvement** and 8.7 **Freshness of Products and Materials**. Additional information is provided in the **Understanding Cybersecurity Risk Freshness in an AVCDL Context** elaboration document.

4.2.11 Industry Best Practices

[G.22] Best practices for secure software development should be followed, for example as outlined in NIST publications and ISO/SAE 21434.

References

ISO/SAE 21434 clause 10 Product Development, Cybersecurity Validation

Discussion

The **AVCDL** was constructed based on available international standards and industry best practice documents. This is detailed in the **AVCDL** primary document section 4 **Background Material**. Additionally, each **AVCDL** document provides a comprehensive list of primary sources upon which it is based.

Note: The **AVCDL** addresses this recommendation far in excess of the scoping and specifics of the provided **ISO/SAE 21434** reference.

[G.23] Manufacturers should actively participate in automotive industry-specific best practices and standards development activities through recognized standards development organizations and Auto-ISAC.

Note: This recommendation is outside the scope of the **AVCDL**.

Discussion

Decisions regarding the participation with external organizations, both standard and regulatory are the sole responsibility of organizational leadership and outside the scope of the **AVCDL**.

[G.24] As future risks emerge; industry should collaborate to expediently develop mitigation measures and best practices to address new risks.

Note: This recommendation is outside the scope of the **AVCDL**.

Discussion

Decisions regarding the participation with external organizations, both standard and regulatory are the sole responsibility of organizational leadership and outside the scope of the **AVCDL**.

4.3 Information Sharing

[G.25] Members of the extended automotive industry (including, but not limited to, vehicle manufacturers, automotive equipment suppliers, software developers, communication services providers, aftermarket system suppliers, and fleet managers) are strongly encouraged to:

[a] Join Auto-ISAC;

Note: This recommendation is outside the scope of the **AVCDL**.

Discussion

Decisions regarding the participation with external organizations, both standard and regulatory are the sole responsibility of organizational leadership and outside the scope of the **AVCDL**.

[b] Share timely information concerning cybersecurity issues, including vulnerabilities, and intelligence information with Auto-ISAC.

Note: This recommendation is outside the scope of the **AVCDL**.

Discussion

Decisions regarding the participation with external organizations, both standard and regulatory are the sole responsibility of organizational leadership and outside the scope of the **AVCDL**.

[G.26] Members of Auto-ISAC are strongly encouraged to collaborate in expeditiously exploring containment options and countermeasures to reported vulnerabilities, regardless of an impact on their own systems.

Note: This recommendation is outside the scope of the **AVCDL**.

Discussion

Decisions regarding the participation with external organizations, both standard and regulatory are the sole responsibility of organizational leadership and outside the scope of the **AVCDL**.

4.4 Security Vulnerability Reporting Program

[G.27] Automotive industry members should create their own vulnerability reporting policies and mechanisms.

Note: This recommendation is partially supported.

References

ISO/SAE 21434 [WP-05-01][RQ-05-02] organization-specific rules / policies

ISO/IEC 29147 Information Technology Security Techniques Vulnerability Disclosure

<https://www.iso.org/standard/72311.html>

Discussion

Vulnerability reporting is an activity led by either the IT or OT group depending on the nature of the incident. As such, product cybersecurity provides support within this context. There are additional coordination elements should the supply chain be involved. The **AVCDL** addresses only those aspects directly related to product cybersecurity.

[Foundation-7]	Incident Response Plan (Foundation-7.2)
[Operation-1]	Cybersecurity Incident Report (Operation-1.1)
[Operation-2]	Cybersecurity Incident Report (Operation-1.1)
[Operation-3]	Cybersecurity Incident Report (Operation-1.1)

4.5 Organizational Incident Response Process

[G.28] Members of the automotive industry should develop a product cybersecurity incident response process. This process should include:

[a] A documented incident response plan;

Note: This recommendation is partially supported.

References

ISO/SAE 21434 [WP-05-01] Cybersecurity policy, rules and processes

ISO/SAE 21434 [WP-08-04] Cybersecurity event assessment

Discussion

Incident response is an activity led by either the IT or OT group depending on the nature of the incident. As such, product cybersecurity provides support within this context. There are additional coordination elements should the supply chain be involved. The **AVCDL** addresses only those aspects directly related to product cybersecurity.

[Foundation-7] Incident Response Plan (Foundation-7.2)

[Operation-1] Cybersecurity Incident Report (Operation-1.1)

[Operation-2] Cybersecurity Incident Report (Operation-1.1)

[Operation-3] Cybersecurity Incident Report (Operation-1.1)

[b] Clearly identified roles and responsibilities within the organization;

Note: This recommendation is partially supported.

References

ISO/SAE 21434 [WP-05-01][RQ-05-03] roles and responsibilities

Discussion

Incident response is an activity led by either the IT or OT group depending on the nature of the incident. As such, product cybersecurity provides support within this context. There are additional coordination elements should the supply chain be involved. The **AVCDL** addresses only those aspects directly related to product cybersecurity.

[Foundation-2] Roles and Responsibilities Document (Foundation-2.1)

[c] Clearly identified communication channels and contacts outside the organization;

Note: This recommendation is partially supported.

References

ISO/SAE 21434 clause 8.3 Cybersecurity Monitoring

ISO/SAE 21434 [WP-08-01][RQ-08-01] cybersecurity monitoring sources

Discussion

Incident response is an activity led by either the IT or OT group depending on the nature of the incident. As such, product cybersecurity provides support within this context. There are additional coordination elements should the supply chain be involved. The **AVCDL** addresses only those aspects directly related to product cybersecurity.

[Foundation-7] [Incident Response Plan \(Foundation-7.2\)](#)

[Operation-1] [Cybersecurity Incident Report \(Operation-1.1\)](#)

[d] Procedures for keeping this information up to date.

Note: This recommendation is partially supported.

Discussion

Incident response is an activity led by either the IT or OT group depending on the nature of the incident. As such, product cybersecurity provides support within this context. There are additional coordination elements should the supply chain be involved. The **AVCDL** addresses only those aspects directly related to product cybersecurity.

[Foundation-7] [Incident Response Plan \(Foundation-7.2\)](#)

[G.29] Organizations should develop metrics to periodically assess the effectiveness of their response process.

Note: This recommendation is outside the scope of the **AVCDL**.

Discussion

Incident response is an activity led by either the IT or OT group depending on the nature of the incident. As such, product cybersecurity provides support within this context. There are additional coordination elements should the supply chain be involved. The **AVCDL** addresses only those metrics directly related to product cybersecurity engineering. Additional information is provided in the **AVCDL** primary document section 8.5 **Metrics**.

[G.30] Organizations should document the details of each identified and reported vulnerability, exploit, or incident applicable to their products.

Note: This recommendation is partially supported.

References

ISO/SAE 21434 clause 8 Continuous Cybersecurity Activities

Discussion

Incident response is an activity led by either the IT or OT group depending on the nature of the incident. As such, product cybersecurity provides support within this context. There are additional coordination elements should the supply chain be involved. The **AVCDL** addresses only those aspects directly related to product cybersecurity.

[Foundation-7]	Incident Response Plan (Foundation-7.2)
[Operation-1]	Cybersecurity Incident Report (Operation-1.1)
[Operation-2]	Cybersecurity Incident Report (Operation-1.1)
[Operation-3]	Cybersecurity Incident Report (Operation-1.1)

[G.31] The nature of the vulnerability and the rationale for how the vulnerability is managed should be documented.

Note: This recommendation is partially supported.

Discussion

Incident response is an activity led by either the IT or OT group depending on the nature of the incident. As such, product cybersecurity provides support within this context. There are additional coordination elements should the supply chain be involved. The **AVCDL** addresses only those aspects directly related to product cybersecurity.

[Foundation-7]	Incident Response Plan (Foundation-7.2)
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[Operation-2]	Cybersecurity Incident Report (Operation-1.1)
[Operation-3]	Cybersecurity Incident Report (Operation-1.1)

[G.32] Commensurate to assessed risks, organizations should have a plan for addressing newly identified vulnerabilities on consumer-owned vehicles in the field, inventories of vehicles built but not yet distributed to dealers, vehicles delivered to dealerships but not yet sold to consumers, as well as future products and vehicles.

Note: This recommendation is partially supported.

Discussion

Incident response is an activity led by either the IT or OT group depending on the nature of the incident. As such, product cybersecurity provides support within this context. There are additional coordination elements should the supply chain be involved. The **AVCDL** addresses only those aspects directly related to product cybersecurity.

[Foundation-7] [Incident Response Plan \(Foundation-7.2\)](#)

[G.33] Any incidents should also be reported to CISA/United States Computer Emergency Readiness Team (US-CERT) in accordance with the US-CERT Federal Incident Notification Guidelines.

Note: This recommendation is partially supported.

Discussion

Incident response is an activity led by either the IT or OT group depending on the nature of the incident. As such, product cybersecurity provides support within this context. There are additional coordination elements should the supply chain be involved. The **AVCDL** addresses only those aspects directly related to product cybersecurity.

[Foundation-7] [Incident Response Plan \(Foundation-7.2\)](#)

[G.34] Industry members should periodically conduct and participate in organized, cyber incident response exercises.

Note: This recommendation is partially supported.

Discussion

Cyber incident response exercises (tabletop exercises) are an activity led by either the IT or OT group depending on the nature of the incident. Although tabletop exercises are outside the scope of the **AVCDL**, it is assumed that product cybersecurity individuals would participate in such activities as is appropriate.

4.6 Self-Auditing

4.6.1 Process Management Documentation

[G.35] The automotive industry should document the details related to their vehicle cybersecurity risk management process to facilitate auditing and accountability.

Note: This recommendation is partially supported.

Discussion

The audit process is the responsibility of the compliance group. The **AVCDL** provides support through its fully documented risk management related processes and guidance for handling of documentation generated as a result of their application.

[Foundation-9] Threat Prioritization Plan (Foundation-9.1)
[Design-4] Ranked / Risked Threat Report (Design-4.2)

[G.36] Further, such documents should be retained through the expected lifespan of the associated product.

Note: This recommendation is partially supported.

Discussion

The QMS and DMS are the responsibility of the quality management group. The RMS is the responsibility of the requirements group. The **AVCDL** provides support through its fully documented processes and guidance for handling of documentation generated as a result of their application. Additionally, the global cybersecurity requirements form the basis for all product cybersecurity requirements and are to be held within the RMS.

[G.37] Documents should follow a robust version control protocol, and should be revised regularly as new information, data, and research results become available.

Note: This recommendation is partially supported.

References

ISO/SAE 21434 [WP-05-03][RQ-05-11] quality management system
ISO/SAE 21434 [WP-06-01][RQ-06-12] work products in QMS

Discussion

The QMS is the responsibility of the quality management group. The **AVCDL** provides support through its fully documented processes and guidance for handling of documentation generated as a result of their application.

4.6.2 Review and Audit

[G.38] The automotive industry should establish procedures for internal review of its management and documentation of cybersecurity-related activities.

Discussion

The **AVCDL** addresses this recommendation in detail. This information is provided in the **AVCDL** primary document sections 5 **Continuous Improvement** and 8.7 **Freshness of Products and Materials**. Additional information is provided in the **Understanding Cybersecurity Risk Freshness in an AVCDL Context** elaboration document.

[G.39] The automotive industry should consider carrying out organizational and product cybersecurity audits annually.

Note: This recommendation is partially supported.

References

ISO/SAE 21434 [WP-05-05][RQ-05-17] cybersecurity audit
ISO/SAE 21434 [WP-05-01][RQ-05-03] roles and responsibilities

Discussion

The audit process is the responsibility of the compliance group. Within the context of the product cybersecurity group, the **AVCDL** provides support through its fully documented risk management related processes and guidance for handling of documentation generated as a result of their application.

5 Education

[G.40] Vehicle manufacturers, suppliers, universities, and other stakeholders should work together to help support educational efforts targeted at workforce development in the field of automotive cybersecurity.

Note: This recommendation is partially supported.

Discussion

The organization's training infrastructure is the responsibility of the training group. Within the context of product cybersecurity, the **AVCDL** provides support training materials created to address specific topics, processes and procedures. This material is aligned to the roles and responsibilities specified within the **AVCDL** primary and secondary documentation.

[Foundation-1]	Training Catalog (Foundation-1.1)
[Foundation-1]	System to Track Training Participation (Foundation-1.2)
[Foundation-2]	Roles and Responsibilities Document (Foundation-2.1)

6 Aftermarket / User-Owned Devices

6.1 Vehicle Manufacturers

[G.41] The automotive industry should consider the risks that could be presented by user-owned or aftermarket devices when connected with vehicle systems and provide reasonable protections.

Discussion

The **AVCDL**'s treatment of threat consideration is source agnostic. All entities which may connect to the product via any path through the attack surface are given the same treatment.

[Foundation-9] Threat Prioritization Plan (Foundation-9.1)
[Design-4] Threat Modeling Report (Design-4.1)
[Design-4] Ranked / Risked Threat Report (Design-4.2)
[Design-4] Threat Report (Design-4.3)

[G.42] Any connection to a third-party device should be authenticated and provided with appropriate limited access.

Note: This recommendation is outside the scope of the AVCDL.

Discussion

Access granted to third-party devices is determined at the organizational level.

6.2 Aftermarket Device Manufacturers

[G.43] Aftermarket device manufacturers should employ strong cybersecurity protections on their products.

Discussion

The **AVCDL** is equally applicable to any part of the supply chain including aftermarket device manufacturers.

7 Serviceability

[G.44] The automotive industry should consider the serviceability of vehicle components and systems by individuals and third parties.

Note: This recommendation is outside the scope of the AVCDL.

Discussion

The vehicle component serviceability is an organizational level responsibility.

[G.45] The automotive industry should provide strong vehicle cybersecurity protections that do not unduly restrict access by alternative third-party repair services authorized by the vehicle owner.

Note: This recommendation is outside the scope of the AVCDL.

Discussion

The specific strength of cybersecurity controls is determined by best practices and ultimately determined at the organizational level.

8 Technical Vehicle Cybersecurity Best Practices

8.1 Developer/Debugging Access in Production Devices

[T.1] Developer-level access should be limited or eliminated if there is no foreseeable operational reason for the continued access to an ECU for deployed units.

Note: This recommendation is outside the scope of the **AVCDL**.

Discussion

Configuration and management of access to the release product by various entities is the responsibility of the production and OT groups.

[T.2] If continued developer-level access is necessary, any developer-level debugging interfaces should be appropriately protected to limit access to authorized privileged users.

Cybersecurity Properties

The following cybersecurity properties are challenged:

- Authorization

Cybersecurity Assets

The following cybersecurity assets may be impacted:

- Executables
- Configuration Data
- Databases
- Unstructured Data
- Credentials
- Logs

Cybersecurity Layers

The following cybersecurity layers may be involved:

- Network
- Protocol
- Application

Cybersecurity Requirements

The following global cybersecurity requirements address this recommendation:

CR014	Processes receiving data crossing a trust boundary shall be sandboxed.
CR015	Processes shall be granted the smallest set of privileges.
CR016	Only authorized entities shall perform control operations on executables.
CR017	Privileged access shall be permitted only for singular, specific, time-limited operations.
CR018	The system shall contain only secure default privileged accounts if a privileged account is necessary.
CR019	Processes shall have unique owners.
CR026	Configuration data modification shall only be made by authorized entities.
CR029	Databases shall be segregated to achieve the principle of least access.
CR031	Unstructured data access shall be granted based on the principle of least privilege.
CR037	Credentials shall be modified only by authorized entities.

Discussion

The **AVCDL** provides a set of applicable global cybersecurity requirements which are applicable in the case of resource access. Configuration of access is the responsibility of the production and OT groups.

Note: These requirements are applied independently and depend upon the context of the communication.

8.2 Cryptographic Techniques and Credentials

[T.3] Cryptographic techniques should be current and non-obsolescent for the intended application.

References

FIPS 140-3 Security Requirements for Cryptographic Modules

<https://doi.org/10.6028/NIST.FIPS.140-3>

Cybersecurity Properties

The following cybersecurity properties are challenged:

- Confidentiality
- Integrity
- Authenticity

Cybersecurity Assets

The following cybersecurity assets may be impacted:

- Executables
- Packets
- Memory
- Hardware

Cybersecurity Layers

The following cybersecurity layers may be involved:

- Physical
- Network
- Protocol
- Application

Cybersecurity Requirements

The following global cybersecurity requirements address this recommendation:

CR001	Persistent storage shall be encrypted.
CR002	Update payloads shall be decrypted at time of use.
CR003	Update payloads shall be encrypted at build time.
CR004	Update payloads shall be stored encrypted.
CR005	Executable integrity shall be cryptographically verified.
CR059	Standard network protocols shall be secured using cybersecurity best practices.
CR054	Communication crossing trust boundaries shall ensure data integrity.
CR056	Communication crossing trust boundaries shall be authenticated.
CR058	Custom protocols shall use current best practices for authentication and key exchange (NIST SP 800-57, 63B, 131 and 133).
CR048	Cryptographic material stored in memory shall be handled appropriately.
CR049	Cryptographic material stored in memory shall be validated prior to use.
CR060	Cryptographic operations shall use a mechanism that is backed by a hardware root of trust.

Discussion

The **AVCDL** global cybersecurity requirements catalog specifies exact, constrained, and testable requirements based on the desire to ensure the confidentiality, integrity, and authenticity of assets (executables, packets, memory, hardware). These considerations may be applied at every layer.

Note: These requirements are applied independently and depend upon the context of the asset under consideration.

[T.4] Cryptographic credentials that provide an authorized, elevated level of access to vehicle computing platforms should be protected from unauthorized disclosure or modification.

Note: This recommendation is supported the **AVCDL**.

Cybersecurity Properties

The following cybersecurity properties are challenged:

- Authorization

Cybersecurity Assets

The following cybersecurity assets may be impacted:

- Credentials

Cybersecurity Layers

The following cybersecurity layers may be involved:

- Application

Cybersecurity Requirements

The following global cybersecurity requirements address this recommendation:

CR037	Credentials shall be modified only by authorized entities.
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Discussion

The responsibility for configuration and management of access control lies with the production and OT groups.

[T.5] Any credential obtained from a single vehicle's computing platform should not provide access to other vehicles.

Note: This recommendation is outside the scope of the **AVCDL**.

Discussion

The management and distribution of cryptographic materials at, or post end-of-line is the responsibility of the production and OT groups.

8.3 Vehicle Diagnostic Functionality

[T.6] Diagnostic features should be limited, as much as possible, to a specific mode of vehicle operation which accomplishes the intended purpose of the associated feature.

Note: This recommendation is outside the scope of the **AVCDL**.

Discussion

The decisions as to the number and scope of diagnostic capabilities available is the responsibility of the systems engineering and development groups.

[T.7] Diagnostic operations should be designed to eliminate or minimize potentially dangerous ramifications if they were misused or abused outside of their intended purposes.

Note: This recommendation is outside the scope of the **AVCDL**.

Discussion

The decisions as to the number and scope of diagnostic capabilities available is the responsibility of the systems engineering and development groups.

Note: It is a general **AVCDL** recommendation that attack surfaces be minimized.

[T.8] The use of global symmetric keys and ad-hoc cryptographic techniques for diagnostic access should be minimized.

Note: This recommendation is supported by the **AVCDL**.

Cybersecurity Properties

The following cybersecurity properties are challenged:

- Integrity
- Authenticity

Cybersecurity Assets

The following cybersecurity assets may be impacted:

- Packets

Cybersecurity Layers

The following cybersecurity layers may be involved:

- Network
- Protocol
- Application

Cybersecurity Requirements

The following global cybersecurity requirements address this recommendation:

CR054	Communication crossing trust boundaries shall ensure data integrity.
CR056	Communication crossing trust boundaries shall be authenticated.
CR058	Custom protocols shall use current best practices for authentication and key exchange (NIST SP 800-57, 63B, 131 and 133).

Discussion

The **AVCDL** provides global cybersecurity requirements for implementation of secure communications, however the development group has the final say as to the specific mechanisms chosen. Additionally, ongoing cryptographic material management is the responsibility of the OT group.

Note: These requirements are applied independently and depend upon the context of the communication.

8.4 Diagnostic Tools

[T.9] Vehicle and diagnostic tool manufacturers should control tools' access to vehicle systems that can perform diagnostic operations and reprogramming by providing for appropriate authentication and access control.

Note: This recommendation is outside the scope of the **AVCDL**.

References

ISO/SAE 21434 [WP-05-04][RQ-05-14] manage tools

Discussion

The selection, configuration, and management of diagnostic tools is the responsibility of the development and OT groups.

Note: The **AVCDL** recommends that all executables and systems use appropriately constrained access mechanisms.

8.5 Vehicle Internal Communications

[T.10] When possible, critical safety signals should be transported in a manner inaccessible through external vehicle interfaces.

Note: This recommendation is outside the scope of the **AVCDL**.

Discussion

Safety-specific design considerations are the responsibility of the safety group.

[T.11] Employ best practices for communication of critical information over shared and possibly insecure channels. Limit the possibility of replay, integrity compromise, and spoofing. Physical and logical access should also be highly restricted.

Cybersecurity Properties

The following cybersecurity properties are challenged:

- Confidentiality
- Integrity

Cybersecurity Assets

The following cybersecurity assets may be impacted:

- Packets

Cybersecurity Layers

The following cybersecurity layers may be involved:

- Network
- Protocol

Cybersecurity Requirements

The following global cybersecurity requirements address this recommendation:

CR059	Standard network protocols shall be secured using cybersecurity best practices.
CR054	Communication crossing trust boundaries shall ensure data integrity.
CR056	Communication crossing trust boundaries shall be authenticated.
CR058	Custom protocols shall use current best practices for authentication and key exchange (NIST SP 800-57, 63B, 131 and 133).

Discussion

The **AVCDL** global security requirements catalog specifies exact, constrained, and testable requirements based on the desire to ensure the confidentiality, integrity, and authenticity of assets (executables, packets, memory, hardware). These considerations may be applied at every layer.

Note: These requirements are applied independently and depend upon the context of the asset under consideration.

Note: Many requirements presume implementation through the application of well-known best practices.

8.6 Event Logs

[T.12] A log of events sufficient to reveal the nature of a cybersecurity attack or successful breach and support event reconstruction should be created and maintained.

Note: This recommendation is outside the scope of the **AVCDL**.

Discussion

The determination of log entry content and sufficiency is the responsibility of development.

Note: It is presumed that the product cybersecurity group would provide guidance where appropriate.

[T.13] Such logs that can be aggregated across vehicles should be periodically reviewed to assess potential trends of cyberattacks.

Note: This recommendation is outside the scope of the **AVCDL**.

Discussion

The management of logs is the responsibility of the OT group.

Note: It is presumed that the product cybersecurity group would provide log interpretation guidance where appropriate.

8.7 Wireless Paths Into Vehicles

8.7.1 Wireless Interfaces

[T.14] Manufacturers should treat all networks and systems external to a vehicle's wireless interfaces as untrusted and use appropriate techniques to mitigate potential threats.

Note: This recommendation is outside the scope of the **AVCDL**.

Discussion

Although the **AVCDL** does not address the topic of wireless communication specifically, all cybersecurity requirements relevant to communication between entities would apply. Additionally, when performing threat modeling, it is accepted practice to treat all entities external to the element under consideration as untrusted unless shown otherwise.

8.7.2 Segmentation and Isolation Techniques in Vehicle Architecture Design

[T.15] Network segmentation and isolation techniques should be used to limit connections between wireless-connected ECUs and low-level vehicle control systems, particularly those controlling safety critical functions, such as braking, steering, propulsion, and power management.

Note: This recommendation is outside the scope of the **AVCDL**.

Discussion

The design, configuration and management of networking hardware within the vehicle is the responsibility of the systems engineering, development, and OT groups.

[T.16] Gateways with strong boundary controls, such as strict whitelist-based filtering of message flows between different network segments, should be used to secure interfaces between networks.

Note: This recommendation is outside the scope of the **AVCDL**.

Discussion

The design, configuration and management of networking hardware within the vehicle is the responsibility of the systems engineering, development, and OT groups.

8.7.3 Network Ports, Protocols, and Services

[T.17] Eliminating unnecessary internet protocol services from production vehicles;

Note: This recommendation is outside the scope of the **AVCDL**.

Discussion

The general choice, implementation, configuration, and management of internet protocol services is the responsibility of the development and OT groups.

Note: It is the general recommendation of the **AVCDL** that only necessary communication mechanisms be employed.

[T.18] Limiting the use of network services on vehicle ECUs to essential functionality only;

Note: This recommendation is outside the scope of the **AVCDL**.

Discussion

The general choice, implementation, configuration, and management of network services is the responsibility of the development and OT groups.

Note: It is the general recommendation of the **AVCDL** that only necessary communication mechanisms be employed.

[T.19] Appropriately protecting services over such ports to limit use to authorized parties.

Note: This recommendation is outside the scope of the **AVCDL**.

Discussion

The design, configuration and management of networking hardware within the vehicle is the responsibility of the systems engineering, development, and OT groups.

Note: It is the general recommendation of the **AVCDL** that only appropriate access controls be employed.

8.7.4 Communication to Back-End Servers

[T.20] Manufacturers should use appropriate encryption and authentication methods in any operational communication between external servers and the vehicle.

Cybersecurity Properties

The following cybersecurity properties are challenged:

- Confidentiality
- Integrity

Cybersecurity Assets

The following cybersecurity assets may be impacted:

- Packets

Cybersecurity Layers

The following cybersecurity layers may be involved:

- Network
- Protocol

Cybersecurity Requirements

The following global cybersecurity requirements address this attack vector:

CR059	Standard network protocols shall be secured using cybersecurity best practices.
CR054	Communication crossing trust boundaries shall ensure data integrity.
CR056	Communication crossing trust boundaries shall be authenticated.
CR058	Custom protocols shall use current best practices for authentication and key exchange (NIST SP 800-57, 63B, 131 and 133).

Discussion

The **AVCDL** global security requirements catalog specifies exact, constrained, and testable requirements based on the desire to ensure the confidentiality, integrity, and authenticity of assets (executables, packets, memory, hardware). These considerations may be applied at every layer.

Note: These requirements are applied independently and depend upon the context of the asset under consideration.

Note: Many requirements presume implementation through the application of well-known best practices.

8.7.5 Capability to Alter Routing Rules

[T.21] Manufacturers should plan for and create processes that could allow for quickly propagating and applying changes in network routing rules to a single vehicle, subsets of vehicles, or all vehicles connected to the network.

Note: This recommendation is outside the scope of the **AVCDL**.

Discussion

The responsibility for configuration, management, and update of networking hardware is the responsibility of the OT group.

8.8 Software Updates / Modifications

[T.22] Automotive manufacturers should employ state-of-the-art techniques for limiting the ability to modify firmware to authorized and appropriately authenticated parties.

Discussion

The **AVCDL** addresses the treatment of software update in the **AVCDL Phase Requirement Product ISO 24089 Work Product Fulfillment Summary** certification document.

[T.23] Manufacturers should employ measures to limit firmware version rollback attacks.

Discussion

The **AVCDL** addresses the treatment of software update in the **AVCDL Phase Requirement Product ISO 24089 Work Product Fulfillment Summary** certification document.

8.9 Over-the-Air Software Updates

[T.24] Maintain the integrity of OTA updates, update servers, the transmission mechanism, and the updating process in general.

Discussion

The **AVCDL** addresses the treatment of software update in the **AVCDL Phase Requirement Product ISO 24089 Work Product Fulfillment Summary** certification document.

[T.25] Take into account, when designing security measures, the risks associated with compromised servers, insider threats, men-in-the-middle attacks, and protocol vulnerabilities.

Discussion

The **AVCDL** addresses the treatment of software update in the **AVCDL Phase Requirement Product ISO 24089 Work Product Fulfillment Summary** certification document.